2.1.

一维概率密度：$m_X(t) = E[Vt+b] = tEV + b = b$

$D_X(t) = D[Vt+b] = t^2 DV + Db = t^2$

$\therefore f(x,t) = \frac{1}{\sqrt{2\pi}t} \exp\left(-\frac{1}{2}\frac{(x-b)^2}{t^2}\right), \ t \in (0,\infty)$

对 $\forall s,t \in T$，$R_X(s,t) = E[X(s)X(t)] = E[(Vs+b)(Vt+b)]$

$= stEV^2 + b^2 = st + b^2$

2.2

设 $y=g(x)$，已知 $f_X(x)$，求 $f_Y(y)$

若 $y=g(x)$ 在 X 的取值范围内严格单调可导，且有反函数 $x=h(y)$

$F_Y(y) = P(Y \le y) = P(h(Y) \le h(y)) = P(X \le h(y)) = F_X(h(y))$

$\therefore f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{dF_X(h(y))}{dy} = f_X(h(y)) \cdot |h'(y)|$

反函数 $y = -\frac{\ln x}{t}$ $\quad f_X(x) = f_Y(-\frac{\ln x}{t}) \cdot |-\frac{1}{xt}|$

$x \in (0,1)$ $\quad = \frac{f(-\frac{\ln x}{t})}{tx}, \ t>0.$

$m_X(t) = EX(t) = E(e^{-Yt}) = \int_0^\infty f(y) e^{-yt} dy$

$R_X(t_1,t_2) = E[X(t_1)X(t_2)] = E(e^{-Y(t_1+t_2)}) = \int_0^\infty e^{-y(t_1+t_2)} f(y) dy$

2.3

(1) $t=\frac{1}{2}$ 时，$X(\frac{1}{2}) = \begin{cases} 0, & 正 \\ 1, & 反 \end{cases}$ $\quad \therefore P(X(\frac{1}{2})=0) = P(X(\frac{1}{2})=1) = \frac{1}{2}$

$F(\frac{1}{2};x) = \begin{cases} 0, & x<0 \\ \frac{1}{2}, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$

同理当 $t=1$，$X(1) = \begin{cases} -1, & 正 \\ 2, & 反 \end{cases}$ $\quad F(1;x) = \begin{cases} 0, & x<-1 \\ \frac{1}{2}, & -1 \le x < 2 \\ 1, & x \ge 2 \end{cases}$

(2) 先找出分布列 $P(X(\frac{1}{2})=0, X(1)=-1) = \frac{1}{4}$

$P(X(\frac{1}{2})=0, X(1)=2) = \frac{1}{4}$

$P(X(\frac{1}{2})=1, X(1)=-1) = \frac{1}{4}$

$P(X(\frac{1}{2})=1, X(1)=2) = \frac{1}{4}$

$F(\frac{1}{2},1;x_1,x_2) = \begin{cases} 0, & x_1<0 \text{ 且 } x_2<-1 \\ \frac{1}{4}, & 0 \le x_1 < 1 \text{ 且 } -1 \le x_2 < 2 \\ \frac{1}{2}, & 0 \le x_1 < 1 \text{ 且 } x_2 \ge 2 \text{ 或 } x_1 \ge 1 \text{ 且 } -1 \le x_2 < 2, \\ 1, & x_1 \ge 1 \text{ 且 } x_2 \ge 2 \end{cases}$

(3) $m_X(t) = EX(t) = \frac{1}{2}\cos(\pi t) + t$

$m_X(1) = \frac{1}{2}$ $\quad \sigma_X^2(t) = E(X(t)-m_X(t))^2 = \frac{1}{2}(\cos(\pi t) - \frac{1}{2}\cos(\pi t) - t)^2 + \frac{1}{2}(2t - \frac{1}{2}\cos(\pi t) - t)^2$

$= \frac{1}{2}(\frac{1}{2}\cos(\pi t) - t)^2 + \frac{1}{2}(t - \frac{1}{2}\cos(\pi t))^2$

$= (\frac{1}{2}\cos(\pi t) - t)^2 \quad \sigma_X^2(1) = \frac{9}{4}$

2.4

$$m_x(t) = EX(t) = E(A\cos(\omega t) + B\sin(\omega t)) = 0$$

对 $\forall\ t_1, t_2 \in T$, $R_x(t_1,t_2) = E(X(t_1)X(t_2)) = E[(A\cos(\omega t_1) + B\sin(\omega t_1))(A\cos(\omega t_2) + B\sin(\omega t_2))]$

$$= \cos(\omega t_1)\cos(\omega t_2)\,EA^2 + AB\cos(\omega t_1)\cancel{\sin(\omega t_2)}\underbrace{EAB}_{0} + \sin(\omega t_1)\cancel{\cos(\omega t_2)}\underbrace{EAB}_{0} + \sin(\omega t_1)\underbrace{\sin(\omega t_2)}_{EB^2}$$

$$= \cos[\omega(t_1 - t_2)]\,\sigma^2$$

2.5

$$m_Y(t) = E(X(t) + \varphi(t)) = m_X(t) + \varphi(t)$$

$$B_Y(t_1,t_2) = E[(Y(t_1) - m_Y(t_1))(Y(t_2) - m_Y(t_2))] = EY(t_1)Y(t_2) - m_Y(t_1)m_Y(t_2)$$

$$= E(X(t_1) + \varphi(t_1))(X(t_2) + \varphi(t_2)) - (m_X(t_1) + \varphi(t_1))(m_X(t_2) + \varphi(t_2))$$

$$= EX(t_1)X(t_2) + \varphi(t_2)EX(t_1) + \varphi(t_1)EX(t_2) + \varphi(t_1)\varphi(t_2) - (m_X(t_1) + \varphi(t_1))(m_X(t_2) + \varphi(t_2))$$

$$= B_X(t_1,t_2) + \cancel{m_X(t_1)m_X(t_2)} + \cancel{\varphi(t_2)m_X(t_1)} + \cancel{\varphi(t_1)m_X(t_2)} + \cancel{\varphi(t_1)\varphi(t_2)}$$
$$- \cancel{m_X(t_1)m_X(t_2)} - \cancel{m_X(t_1)\varphi(t_2)} - \cancel{\varphi(t_1)m_X(t_2)} - \cancel{\varphi(t_1)\varphi(t_2)}$$

$$= B_X(t_1,t_2)$$

2.6

$$R_Y(t,t+\tau) = EY(t)Y(t+\tau) = E[A^2\sin^2(\omega t+\theta)\cdot A^2\sin^2(\omega(t+\tau)+\theta)]$$

$$= \int_{-\pi}^{\pi} \frac{A^4}{2\pi}\sin^2(\omega t+\theta)\sin^2(\omega(t+\tau)+\theta)\,d\theta$$

$$= \frac{A^4}{2\pi}\cdot\frac{1}{4}\int_{-\pi}^{\pi}(1 - \cos(2\omega t + 2\omega\tau + 2\theta) - \cos(2\omega t + 2\theta) +$$
$$\cos(2\omega t + 2\theta)\cos(2\omega t + 2\omega\tau + 2\theta))\,d\theta$$

$$\int_{-\pi}^{\pi}\cos(2\omega t + 2\omega\tau + 2\theta)\,d\theta = \frac{1}{2}\left(\sin(2\theta + 2\omega t + 2\omega\tau)\Big|_{\theta=\pi} - \sin(2\theta + 2\omega t + 2\omega\tau)\Big|_{\theta=-\pi}\right)$$

$$= \frac{1}{2}(\sin(2\omega t + 2\omega\tau) - \sin(2\omega t + 2\omega\tau)) = 0$$

$$\therefore R_Y(t,t+\tau) = \frac{A^4}{8\pi}\int_{-\pi}^{\pi} 1 - \cos(2\omega t + 2\theta) + \cos(2\omega t + 2\theta)\cos(2\omega t + 2\omega\tau + 2\theta)\,d\theta$$

$$= \frac{1}{4}A^4 + \frac{A^4}{8\pi}\int_{-\pi}^{\pi}\cos(2\omega t + 2\theta)\cos(2\omega t + 2\omega\tau + 2\theta)\,d\theta$$

$$\cos(2\omega t + 2\theta)(\cos(2\omega t + 2\theta)\cos 2\omega\tau - \sin(2\omega t + 2\theta)\sin 2\omega\tau)$$

$$\int_{-\pi}^{\pi}[\cos 2\omega\tau\cos^2(2\omega t + 2\theta) - \sin 2\omega\tau\sin(2\omega t + 2\theta)\cos(2\omega t + 2\theta)]\,d\theta$$

$$\int_{-\pi}^{\pi}\cos 2\omega\tau\left(\frac{1 + \cos(4\omega t + 4\theta)}{2}\right)d\theta = \pi\cdot\cos 2\omega\tau$$

$$\therefore R_Y(t,t+\tau) = \frac{1}{4}A^4 + \frac{A^4}{8\pi}\cdot\pi\cos 2\omega\tau = \frac{1}{4}A^4\left(1 + \frac{1}{2}\cos 2\omega\tau\right)$$

$$R_{XY}(t,t+\tau) = E(X(t)Y(t+\tau)) = E(A\sin(\omega t+\theta)\cdot A^2\sin^2(\omega t + \omega\tau + \theta))$$

$$= \frac{A^3}{2\pi}\int_{-\pi}^{\pi} \sin(wt+\theta)\sin^2(wt+w\tau+\theta)\,d\theta = 0$$

2.7

$B_X(t_1,t_2) = R_X(t_1,t_2) - m_X(t_1)m_X(t_2)$ $\quad DX = DY = DZ = 1$

$\because R_X(t_1,t_2) = E(X+Yt_1+Zt_1^2)(X+Yt_2+Zt_2^2) = EX^2 + t_2\cancel{EXY} + t_2^2\cancel{EXZ} + t_1\cancel{EXY} + t_1t_2EY^2 + t_1t_2^2\cancel{EYZ}$

$+ t_1^2\cancel{EXZ} + t_1^2t_2\cancel{EZY} + t_1^2t_2^2EZ^2$

$= 1 + t_1t_2 + t_1^2t_2^2$

$m_X(t) = E(X+Yt+Zt^2) = 0$

$\therefore B_X(t_1,t_2) = t_1^2t_2^2 + t_1t_2 + 1$

2.8

$X(t)$的一维分布函数可写为 $F_X(x,t) = P(X(t)\le x)$

$m_Y(t) = EY(t) = P(Y(t)=y_1)y_1 + P(Y(t)=y_2)y_2$，其中 $y_1=1$，$y_2=0$

$\therefore EY(t) = P(Y(t)=1) = P(X(t)\le x) = F_X(x,t)$

$R_Y(t_1,t_2) = E[Y(t_1)Y(t_2)] = 1\cdot1\cdot P\{X(t_1)\le x_1, X(t_2)\le x_2\} + 1\cdot0\cdot P\{X(t_1)\le x_1, X(t_2)>x_2\}$

$+ 0\cdot1\cdot P\{X(t_1)>x_1, X(t_2)\le x_2\} + 0\cdot0\cdot P\{X(t_1)>x_1, X(t_2)>x_2\}$

$= P\{X(t_1)\le x_1, X(t_2)\le x_2\}$

2.9

$E[X(t)X(t+\tau)] = E[f(t-Y)f(t+\tau-Y)] = \int_0^T f(t-y)f(t+\tau-y)f_Y(y)\,dy \quad f_Y(y)=\frac{1}{T}$

令 $s=t-y$，$\frac{1}{T}\int_0^T f(s)f(s+\tau)\,d(t-s) = -\frac{1}{T}\int_t^{t-T} f(s)f(s+\tau)\,ds = \frac{1}{T}\int_{t-T}^{t} f(s)f(s+\tau)\,ds.$

$= \frac{1}{T}\int_0^T f(s)f(s+\tau)\,ds.$

2.10

(1) $|B_X(t_1,t_2)| = |R_X(t_1,t_2) - m_X(t_1)m_X(t_2)|$ (Schwarz不等式)

$\because \sigma_X^2(t_1)\sigma_X^2(t_2) = (EX(t_1)^2 - m_X(t_1)^2)(EX(t_2)^2 - m_X(t_2)^2)$

$= EX(t_1)^2\cdot EX(t_2)^2 - EX(t_1)^2m_X(t_2)^2 - m_X(t_1)^2EX(t_2)^2 + m_X(t_1)^2m_X(t_2)^2$

$\therefore |B_X(t_1,t_2)|^2 = E(X_1)^2\cdot EX(t_2)^2 - 2EX(t_1)EX(t_2)m_X(t_1)m_X(t_2) + m_X(t_1)^2m_X(t_2)^2$

又 $(EX(t_1)m_X(t_2) - EX(t_2)m_X(t_2))^2 \ge 0$ $\quad$ 得证 $|B_X(t_1,t_2)| \le \sigma_X(t_1)\sigma_X(t_2)$

(2) 即证 $\sigma_X(t_1)\sigma_X(t_2) \le \frac{1}{2}[\sigma_X^2(t_1) + \sigma_X^2(t_2)]$ $\quad \because (\sigma_X(t_1)-\sigma_X(t_2))^2 \ge 0$ 显然

$|B_X(t_1,t_2)| \le \sigma_X(t_1)\sigma_X(t_2) \le \frac{1}{2}[\sigma_X^2(t_1) + \sigma_X^2(t_2)]$

2.11 同样由Schwarz不等式证明

2.12

$$m_X(t) = E\left[\sum_{k=1}^{N} A_k e^{i(wt+\phi_k)}\right] = \sum_{k=1}^{N} E\left[A_k e^{i(wt+\phi_k)}\right] = \sum_{k=1}^{N} E\left[A_k(\cos(wt+\phi_k) + i\sin(wt+\phi_k))\right]$$

$$= \sum_{k=1}^{N} E\left[A_k \cos(wt+\phi_k)\right] + i\sum_{k=1}^{N} E\left[A_k \sin(wt+\phi_k)\right]$$

$$= \sum_{k=1}^{N} EA_k \cdot E\cos(wt+\phi_k) + i\sum_{k=1}^{N} EA_k \cdot E\sin(wt+\phi_k)$$

其中，$E\cos(wt+\phi_k) = \int_0^{2\pi} \frac{1}{2\pi}\cos(wt+\phi_k)\,d\phi_k = 0$ $\quad E\sin(wt+\phi_k) = 0$

$\therefore m_X(t) = 0$

$$B_X(t_1,t_2) = R_X(t_1,t_2) = E\,X(t_1)\overline{X(t_2)} = E\left[\sum_{k=1}^{N} A_k e^{i(wt_1+\phi_k)} \cdot \sum_{j=1}^{N} A_j e^{-i(wt_2+\phi_j)}\right]$$

$$= \sum_{k=1}^{N} E\left[A_k e^{i(wt_1+\phi_k)} \cdot \sum_{j=1}^{N} A_j e^{-i(wt_2+\phi_j)}\right]$$

$$= \sum_{k=1}^{N}\sum_{j=1}^{N} E\left[A_k A_j e^{i(wt_1+\phi_k)} e^{-i(wt_2+\phi_j)}\right]$$

$$= \sum_{k=1}^{N}\sum_{j=1}^{N} E\left[A_k A_j\right] \cdot E e^{i[w(t_1-t_2)+(\phi_k-\phi_j)]}$$

当$k \neq j$，ϕ_k与ϕ_j独立，$E e^{i[w(t_1-t_2)+(\phi_k-\phi_j)]} = e^{iw(t_1-t_2)} \cdot E e^{i\phi_k} \cdot E e^{-i\phi_j} = 0$

当$k=j$，$B_X(t_1,t_2) = e^{iw(t_1-t_2)} \cdot \sum_{k=1}^{N} EA_k^2$

2.15